

# Botox Markdown Suite

*Native LaTeX-Grade Typesetting & Slide Decks in a Single Binary*

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**Abstract** – *Botox is a pure Rust, single-binary command-line suite that compiles standard Markdown documents into publication-grade LaTeX-style PDF documentation or presentation slide decks. Featuring an embedded in-process typesetting engine, authentic New Computer Modern typography, advanced TeX mathematics, native IEEE-standard bibliographies, and Pandoc extensions, Botox requires zero external runtime dependencies.*

**Keywords** – *Document Engineering, LaTeX Typesetting, Pandoc Compatibility, Rust*

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# 1 Overview and Architecture

Botox provides a single, self-contained native executable ( `botox` ) that transforms standard Markdown files into either publication-grade LaTeX-style PDF documents or clean 16:9 presentation slide decks.

Mode detection is automatic based on frontmatter structure (or can be explicitly declared with `--pdf` or `--slides` ). This `README.md` file itself serves as a comprehensive living specification of all supported features and can be compiled directly:

```
botox README.md -o README.pdf
```

## 1.1 Architectural Pillars

Unlike traditional document compilation pipelines that require a web of external runtimes:

- **Zero Runtime Dependencies:** No Node.js, no Python, no Pandoc runtime, no headless Chrome, and no multi-gigabyte TeX Live distributions.
- **Embedded Typesetting Engine:** Built on the [Rust Language](#) [1] with the [Typst](#) [2] layout engine and New Computer Modern font family (Roman, Bold, Italic, and Math) compiled directly into the binary<sup>1</sup>.
- **Instantaneous Compilation:** Sub-second compilation times (typically 400ms to 900ms) for multi-page documents.
- **Cross-Platform Portability:** Builds as a single static executable for Linux, macOS, and Windows.

## 1.2 Feature Matrix

- [x] Standalone native single binary ( `botox` ) with zero external requirements
- [x] Complete Pandoc frontmatter compatibility ( `geometry` , `lang` , `toc-depth` , `number-sections` )
- [x] Typography extensions: subscripts (  $H_{2-0}$  ), superscripts (  $10^6^$  ), strikethrough ( `~~text~~` )
- [x] Native multi-pass footnotes with bottom-of-page rule separation
- [x] Definition lists and GFM task checklists
- [x] Advanced TeX mathematical notation (matrices, piecewise systems, partial derivatives, accents)
- [x] Pipe tables with column alignments ( `:---` , `:---:` , `---:` ) and cell formatting
- [x] Automatic IEEE-standard numbered bibliography generated from hyperlinks
- [x] High-resolution figures and vector graphics (PNG, JPEG, SVG, WebP, GIF) with automated captions and numbering
- [x] Marp-compatible 16:9 presentation slide decks partitioned by `---`
- [x] Cascading configuration hierarchy ( `botox.yaml` )

---

<sup>1</sup>Embedded font assets include 17 New Computer Modern and DejaVu typeface variants loaded directly from in-memory binary slices.

## 2 Installation and Quick Start

### 2.1 System Binary Installation

The pre-compiled binary can be copied directly to any folder on your system `PATH` :

```
# Install to user local binary path
cp bin/botox ~/.local/bin/botox
```

### 2.2 Building from Source

To compile the standalone binary from source using the Rust toolchain:

```
cd botox
cargo build --release
strip target/release/botox
cp target/release/botox ~/.local/bin/botox
```

### 2.3 Unified CLI Invocation

The single executable `botox` handles both documents and presentations:

```
# Initialize a new document or presentation with active default settings
botox init document.md
botox init slides.md

# Compile documentation PDF (auto-detected)
botox README.md -o README.pdf

# Compile presentation slides (auto-detected)
botox slides.md -o slides.pdf

# Transform hyperlinks into an IEEE Bibliography
botox document.md --bib -o document.pdf

# Toggle automatic Table of Contents
botox document.md --toc -o document.pdf
botox document.md --no-toc -o document.pdf

# Inspect active configuration settings
botox config
```

## 3 Typography and Pandoc Extensions

Botox adheres to standard Pandoc Markdown conventions and extensions.

### 3.1 Inline Styling and Formatting

Standard emphasis, bold, code, and strikethrough are fully supported alongside intra-word subscript and superscript delimiters:

- **Chemical Formulas & Exponents:** Water is written as  $\text{H}_2\text{O}$ , glucose as  $\text{C}_6\text{H}_{12}\text{O}_6$ , and astronomical orders of magnitude as  $10^6$  or  $10^{-19}$ .

- **Strikethrough:** Deprecated workflows are marked with `legacy-pipelines` and replaced with instant native compilation.
- **Code Spans:** Variables and registers like `reg_ex_mem` and `instruction_pointer` are rendered with monospace background badges.

### 3.2 Definition Lists

Pandoc-style definition lists format technical glossaries with bold headwords and indented definitions:

**Zero-Dependency Architecture** A self-contained executable that bundles all fonts, engines, and converters without external interpreters.

**Forwarding Unit** A microarchitectural hazard-mitigation bypass that routes execution results directly between pipeline stages.

**Automatic Bibliography** In-text citations and end-of-document reference sections generated automatically from standard Markdown links.

### 3.3 Blockquotes

Academic citations and callouts render with subtle left border rules and margins:

““Simplicity is prerequisite for reliability.” — Edsger W. Dijkstra, *Selected Writings on Computing*”

### 3.4 Callout Divs

Pandoc-style fenced divs ( `:::` ) render as color-coded advisory blocks with distinct visual borders:

#### Note

**Pandoc `link_attributes`:** The `{width=50%}` syntax originates from Pandoc’s `link_attributes` extension. Botox natively parses and renders image attributes without requiring any external preprocessors.

#### Tip

Use cross-references like `@fig:arch`, `@tbl:perf`, `@sec:math`, or `@eq:energy` to generate clickable hyperlinks that automatically track section and asset numbering.

#### Warning

Never mix unnumbered headings with strict numerical cross-references. Unnumbered headings ( `{-}` or `{.unnumbered}` ) are excluded from the numbering counter.

### 3.5 Figures, Graphics & Attributes

Raster and vector image assets (PNG, JPEG, SVG, WebP, GIF) are rendered natively without external decoders. Standard Markdown syntax with alt text generates numbered LaTeX figures with captions:

```
![Botox Architecture Pipeline](figures/architecture.png){width=65% #fig:arch}  
{width=10cm height=5cm}  
![Relative Scaling](banner.png){width=0.8\linewidth}
```

Attributes inside `{...}` support:

- Dimensions: `%` (e.g. `width=50%`), absolute units (`10cm`, `4in`, `200pt`, `100mm`), pixels (`300px`), and TeX factors (`0.8\linewidth`, `\textwidth`).
- Identifiers: `#fig:id` or `id=fig:id` for cross-referencing with `@fig:id`.

### 3.6 Cross-Referencing

Botox implements Pandoc cross-referencing for figures, tables, sections, and equations:

- `@fig:arch` references Figure 1
- `@tbl:perf` references Table 1 (see Table 1)
- `@sec:math` references Section 4 (see Section 4)
- `@eq:energy` references Equation 1 (see Equation 1)

### 3.7 Page Breaks

LaTeX and Pandoc page break commands placed anywhere in the document are translated into native page divisions:

- `\newpage`
- `\pagebreak`
- `\clearpage`
- `<!-- pagebreak -->`

## 4 Advanced Mathematical Typography

Mathematical expressions render with full LaTeX fidelity using embedded New Computer Modern Math fonts.

### 4.1 Fractions, Accents, and Operators

Nested braces in fractions and roots resolve recursively with proper font scaling:

$$E = mc^2 + \frac{1}{2}mv^2 \tag{1}$$

$$\vec{F} \approx m \frac{\partial^2 \vec{r}}{\partial t^2} + \sqrt[3]{\frac{\alpha_{11}}{\beta + \gamma}} \pm \vec{\varepsilon} \tag{2}$$

Definite integrals, infinite series, and limits preserve standard limits and display styles:

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}, \quad \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x, \quad \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1 \quad (3)$$

## 4.2 Matrix Environments

Parenthesized (`pmatrix`), bracketed (`bmatrix`), and determinant (`vmatrix`) matrices are fully supported:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \left| \begin{array}{cc} \lambda - a & b \\ c & \lambda - d \end{array} \right| \quad (4)$$

## 4.3 Piecewise Cases

Piecewise function definitions with conditions and text descriptions render using the `cases` environment:

$$f(x) = \begin{cases} \sqrt{x} & \text{if } x \geq 0 \\ -\sqrt{-x} & \text{if } x < 0 \end{cases} \quad (5)$$

## 5 Structured Tables

Pipe tables support header alignment markers (`:---` for left, `:---` for center, and `---` for right), automated captions, and cross-reference labels:

| Tool                              | Pipeline Stage | Target Output   | Performance |
|-----------------------------------|----------------|-----------------|-------------|
| Botox Native Engine               | Typst Layout   | PDF Vector      | < 1.0 s     |
| <a href="#">Rust Compiler [1]</a> | Codegen        | Native ELF / PE | $O(N)$      |
| <a href="#">LLVM Backend [3]</a>  | Optimization   | Machine Code    | Parallel    |
| <a href="#">Typst Engine [2]</a>  | Typesetting    | PDF / SVG       | Real-time   |

Table 1: Pipeline execution comparison across toolchains.

Cells seamlessly support inline code, formatting, mathematics, and web links.

## 6 Automatic Bibliography Transformation

When `bibliography: true` is set in the frontmatter (or enabled via the `-b / --bibliography` command-line flag), all standard web links are transformed into an **IEEE-standard numbered bibliography**:

- **In-Text Citations:** Hyperlinks in the body receive bracketed citations (e.g. `[1]`, `[2]`). Clicking the citation number jumps directly to the corresponding entry in the bibliography.
- **Deduplication:** Repeated links to the same destination automatically share the same citation number across the entire document.

- **References Section:** An unnumbered References section is generated at the end of the document featuring hanging indents and electronic resource citations (e.g. including the [GitHub Platform \[4\]](#)).
- **Localization:** Respects the document `lang` parameter (e.g., *References*, *Références*, or *Literaturverzeichnis*).

## 7 Presentation Slide Decks

Slide decks are authored in the same Markdown files, using horizontal rules (---) as slide boundaries. When `marp: true` or slide themes are detected in the frontmatter, Botox compiles the document into a 16:9 presentation PDF:

```
---
marp: true
theme: default
paginate: true
backgroundColor: "#f8fafc"
color: "#0f172a"
---

# Microarchitecture Hazards
Eliott JAQUIER, Shanshe GUNDISHVILI

---

# Forwarding Unit Implementation
- Direct bypass from EX/MEM to ID/EX
- Zero stall cycles for ALU dependencies
- Fallback stall insertion for Load-Use hazards
```

Built-in slide themes include `default`, `academic`, `dark`, and `nord`, with optional custom `backgroundColor` and `color` hex overrides.

## 8 Configuration Reference

Botox loads configuration settings from a cascading hierarchy:

- CLI argument `--config <path>`
- Project-level configuration file: `./botox.yaml`
- User-level global configuration file: `~/.config/botox/config.yaml`

```
document:
  author: "Minus"
  affiliation: "Systems Engineering"
  fontsize: "11pt"
  mainfont: "New Computer Modern"
  mathfont: "New Computer Modern Math"
  papersize: "a4"
  columns: 1
  margin:
```

```

x: "2.5cm"
y: "2.5cm"
section_numbering: true
toc: true
bibliography: true

slides:
  theme: "default"
  paginate: true

```

## 8.1 CLI Option Summary

| Flag                               | Parameter                     | Description                                                 |
|------------------------------------|-------------------------------|-------------------------------------------------------------|
| <code>-o, --output</code>          | <code>&lt;file&gt;</code>     | Output PDF path (default: <code>&lt;input&gt;.pdf</code> )  |
| <code>-b, --bib</code>             | None                          | Enable automatic IEEE Bibliography generation               |
| <code>--no-bib</code>              | None                          | Disable automatic Bibliography generation                   |
| <code>--toc</code>                 | None                          | Force generation of Table of Contents                       |
| <code>--no-toc</code>              | None                          | Disable Table of Contents                                   |
| <code>--author</code>              | <code>&lt;name&gt;</code>     | Override author name                                        |
| <code>--font</code>                | <code>&lt;name&gt;</code>     | Override primary typeface                                   |
| <code>-N, --number-sections</code> | None                          | Enable numbered section headings                            |
| <code>--pdf</code>                 | None                          | Force PDF documentation mode                                |
| <code>--slides</code>              | None                          | Force presentation slide deck mode                          |
| <code>--config</code>              | <code>&lt;file&gt;</code>     | Supply custom configuration YAML path                       |
| <code>init</code>                  | <code>&lt;filename&gt;</code> | Initialize new document or slide deck with default settings |
| <code>config</code>                | Subcommand                    | Display currently active configuration                      |

## Appendix

This appendix section demonstrates an unnumbered heading created with Pandoc’s `{-}` attribute syntax. It appears in the document structure without incrementing the section counter.

## References

- [1] “Rust Language”, [Online]. Available: <https://www.rust-lang.org>.
- [2] “Typst”, [Online]. Available: <https://typst.app>.

- [3] “LLVM Backend”, [Online]. Available: <https://llvm.org>.
- [4] “GitHub Platform”, [Online]. Available: <https://github.com>.